T9.1: (a) $f:[a,b]\longrightarrow\mathbb{R}$ stetig diff'bar.

Ges: $\int_a^b f(x)\, f'(x)\,dx \overset{(*)}{=} \int_a^b f(x)\, d(f(x))$; $y=f(x)$

$f'(x)\,dx = d(f(x)) = f'(x)\,dx$

$(*) = \int_{f(a)}^{f(b)} y\,dy = \frac{1}{2} y^2 \Big|_{f(a)}^{f(b)} = \frac{1}{2}\left[(f(b))^2-(f(a))^2\right]$

$\int \vec{v} = \begin{pmatrix}\int v_1\\ \int v_2\\ \int v_3\end{pmatrix}$

(b) $\int_a^b f(cx+d)\,\underline{dx} = \frac{1}{c}\int_a^b f(cx+d)\; \underset{c\cdot dx}{d(cx+d)} \overset{\xi(x)=cx+d}{=} \frac{1}{c}\int_{\xi(a)}^{\xi(b)} f(\xi)\,d\xi$, Stammfkt von f ist F.

$= \frac{1}{c}\left[F(\xi(b))-F(\xi(a))\right] = \frac{1}{c}\left[F(cb+d)-F(ca+d)\right]$

(c) $\int \frac{\ln x}{x}\,dx$; $f(x)=\ln x$; $f'(x)=\frac{1}{x}$

$= \frac{1}{2}\left[\ln x\right]^2 (+C)$

$x\longmapsto\sqrt{x}$, $c=3$, $d=4$

$\int\sqrt{3x+4}\,dx = \frac{1}{3}\left[\frac{2}{3}(3x+4)^{3/2}\right] = \frac{2}{9}(3x+4)^{3/2}$

$\frac{1}{3}\int\sqrt{3x+4}\;d(3x+4)$

2I

T9.2: (a) $x\longmapsto\frac{1}{\cos^2 x}$, $\tan'(x)=\frac{1}{\cos^2(x)}$

$\int\frac{1}{\cos^2(x)}\,dx = \tan(x)$; auf $\left(k\pi-\frac{\pi}{2},\, k\pi+\pi/2\right)$, $k\in\mathbb{Z}$.

(b) $x\longmapsto\tan(x)$

$\int\tan x\,dx = \int\frac{\sin x}{\cos x}\,dx = \int-\frac{\cos'(x)\,dx}{\cos(x)} = -\int\frac{d(\cos x)}{\cos x} \overset{y=\cos x}{=} -\int\frac{dy}{y}$

$-\ln|\cos x| = -\ln|y|$

(c) $\arctan'(x)=\frac{1}{1+x^2}$

$\int\frac{1}{1+x^2}\,dx = \arctan(x).$

(d) $\int 1\cdot\arctan(x)\,dx = x\arctan(x)-\int\frac{x}{1+x^2}\,dx$ $\quad d(x^2)=2x$

$= x\arctan(x)-\frac{1}{2}\int\frac{d(1+x^2)}{1+x^2} = x\arctan x-\frac{1}{2}\ln|1+x^2|$

(e) $\underbrace{\int\sin^2 x\,dx}_{I} = \int\sin x\cdot\sin(x)\,dx = -\cos x\sin x+\int\cos^2 x\,dx$

$-\cos x\sin x+\int(1-\sin^2 x)\,dx$

$= -\cos x\sin x + x - \underbrace{\int\sin^2 x\,dx}_{I}$

$2I = x-\cos x\sin x$

$I = \frac{1}{2}(x-\cos x\sin x)$

T9.3: z.z.: $\int_0^1 x^n(1-x)^m\,dx = \frac{n!\;m!}{(n+m+1)!}$; $m, n \in \mathbb{N}_0$

→ Mittels vollständige Induktion nach n.

$(1)-(0)$

→ Induktionsanfang: (n = 0)

$$\int_0^1 x^0 (1-x)^m\,dx = \frac{-1}{m+1}\cdot(1-x)^{m+1}\Big|_0^1 = \frac{1}{m+1} \checkmark$$

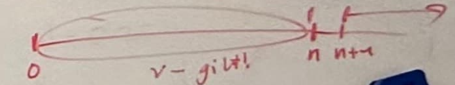
$$\frac{0!\;m!}{(0+m+1)!} = \frac{m!}{(m+1)!} = \frac{1}{m+2}$$

→ I.V.: $\int_0^1 x^\nu (1-x)^m\,dx = \frac{\nu!\;m!}{(\nu+m+1)!}$ $\forall \nu \leq n$, $\forall m \in \mathbb{N}_0$

$$I = \;\; -\frac{1}{2}\int \frac{1}{\sin 2x}\,d(2x) = \frac{1}{2}$$

$\sin(x) = 2\sin\frac{x}{2}\cos\frac{x}{2}$

I.S.: $n \rightsquigarrow n+1$



$$\int x^{n+1}(1-x)^m\,dx$$

P.I.

$$= \underbrace{\left[-\frac{1}{m+1}\cdot x^{n+1}(1-x)^{m+1}\right]_0^1}_{=0} + \frac{n+1}{m+1}\int (1-x)^{m+1}\, x^n\,dx$$

$$= \frac{n+1}{m+1}\int x^n (1-x)^{m+1 = \mu}\,dx$$

$n!\,\mu!$

I.V. $(\nu = n)$

$$= \frac{n+1}{m+2}\cdot\frac{n!\,(m+1)!}{(n+(m+1)+1)!} = \frac{(n+1)!\;m!}{((n+1)+m+1)!}$$

$m!$